- **High-Z Output:** The final output is taken directly from the Op-Amp's output terminal, satisfying the high-impedance (High-Z) requirement and ensuring the waveform remains stable when connected to external loads.

3.8.4 Hardware-Level Calibration

To account for the natural attenuation of the RC filters at higher frequencies (where the signal approaches the f_c), the hardware is supported by a frequency-dependent calibration table. As the user increases the frequency, the software automatically adjusts the MCP4131 wiper to a higher setting to compensate for the hardware's gain roll-off, maintaining a consistent 10V peak-to-peak output across the entire 1–10 kHz range.

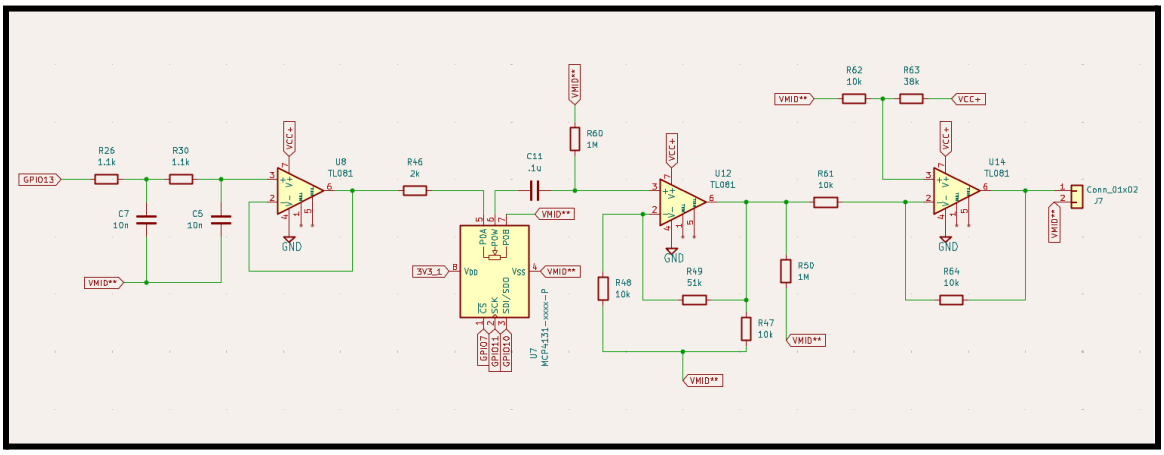


Figure 3.11 - Function Generator Schematic